Card-Fraud Triage

Unsupervised anomaly detection with Gaussian Mixture Models

PHASE 1 · PROJECT 26 · GMM DENSITY SCORING

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Somraj Nandi — Evaluation & Decision Systems



Accuracy hides the fraud problem

0.172%

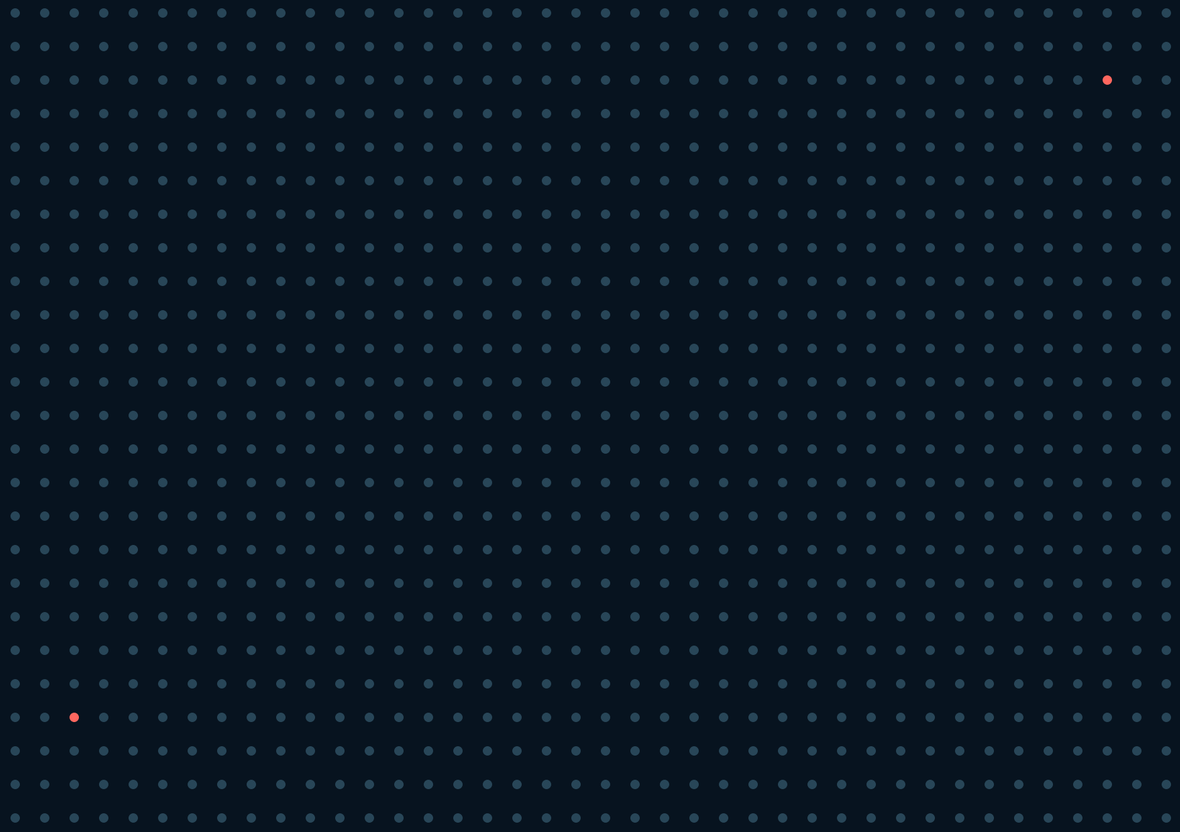
fraud prevalence

284,807 transactions

284,315 legitimate · 492 fraudulent

99.828% accuracy
0 fraud detected

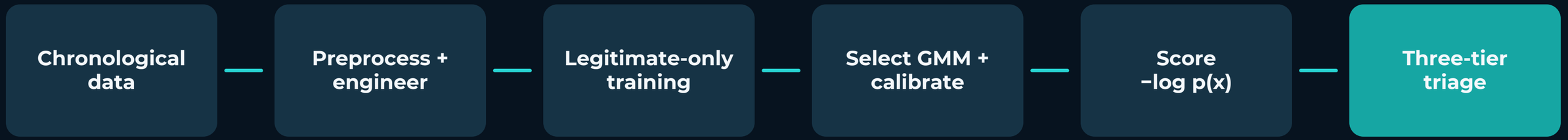
A naive all-legitimate classifier looks accurate—and fails completely.



ILLUSTRATIVE · 2 IN 1,000 HIGHLIGHTED

From density learning to triage decisions

Higher negative log-likelihood → more anomalous



VALIDATION-ONLY CALIBRATION

Untouched test set stays locked until final evaluation.

Chronology prevents lookahead leakage

Sorted by Time · fraud excluded only from model fitting · thresholds frozen before test



198,242 legitimate used

55 fraud

52 fraud

FIT

SELECT + CALIBRATE

EVALUATE ONCE

Robust preprocessing improves rare-event ranking

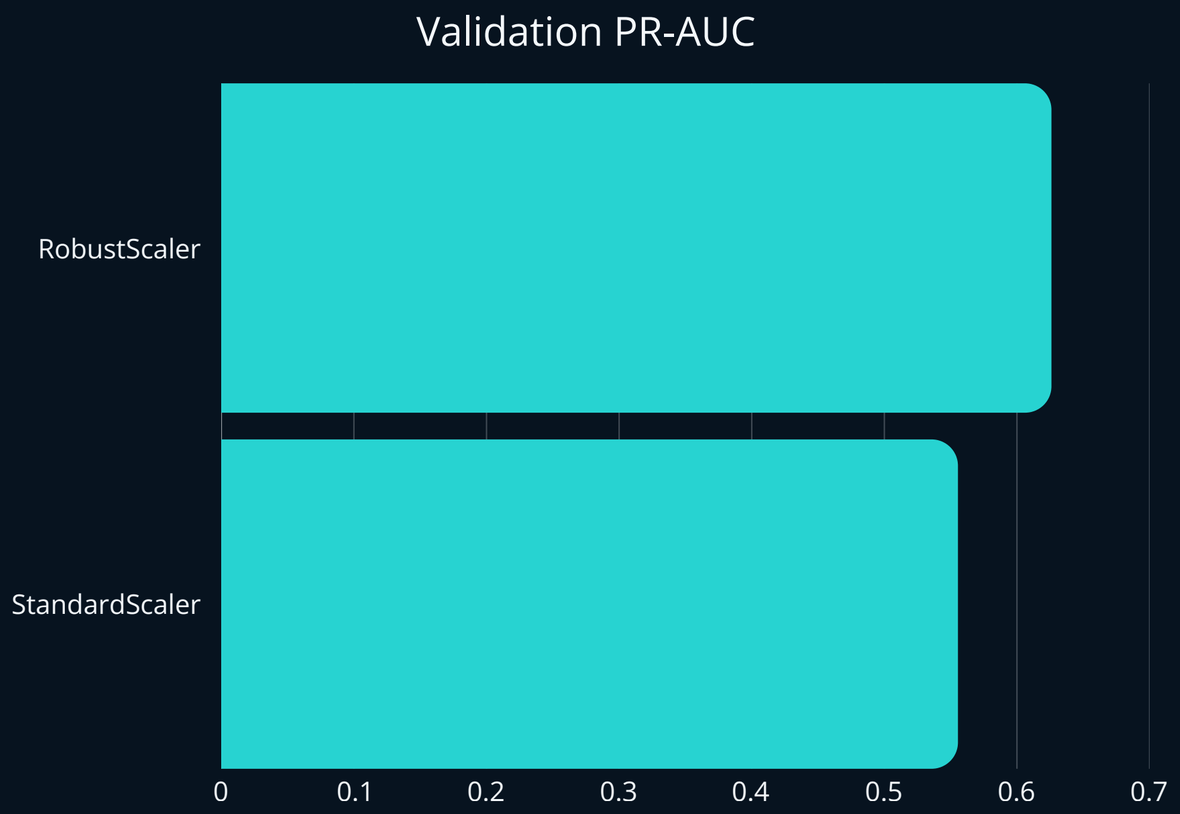
29-D INPUT

V1–V28 + Amount_log

```
Amount_log = ln(1 + Amount)
```

Compresses the heavy tail and handles zero-value amounts gracefully.

RobustScaler fitted strictly on legitimate training data



GMM learns the shape of legitimate behavior

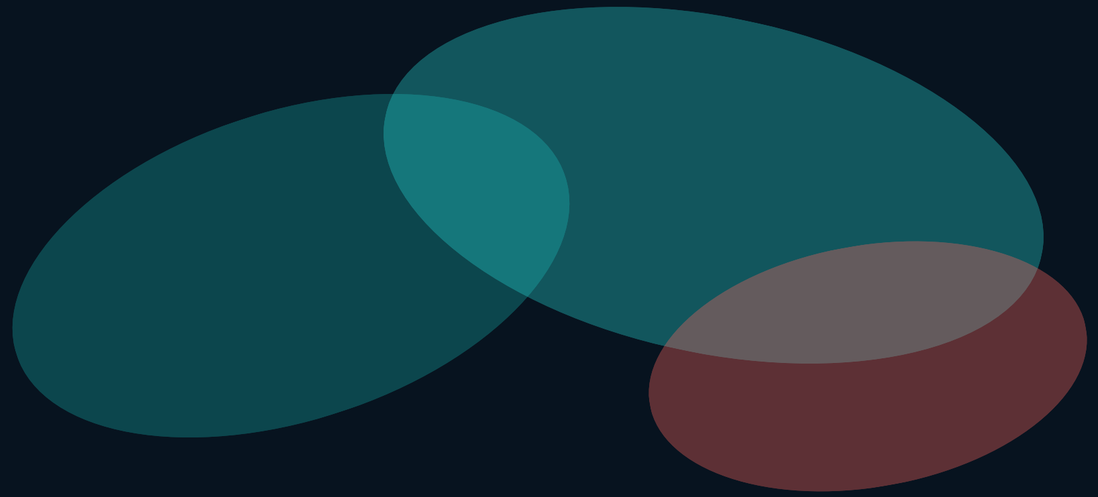
$$p(x|\theta) = \Sigma\ \pi_k\ \mathcal{N}(x|\mu_k,\ \Sigma_k)$$

PRODUCTION CONFIGURATION
K = 14 · full covariance · 29 dimensions
reg_covar = 10^{-6} · 198,242 legitimate tx

E-STEP
responsibilities

—

M-STEP
update parameters



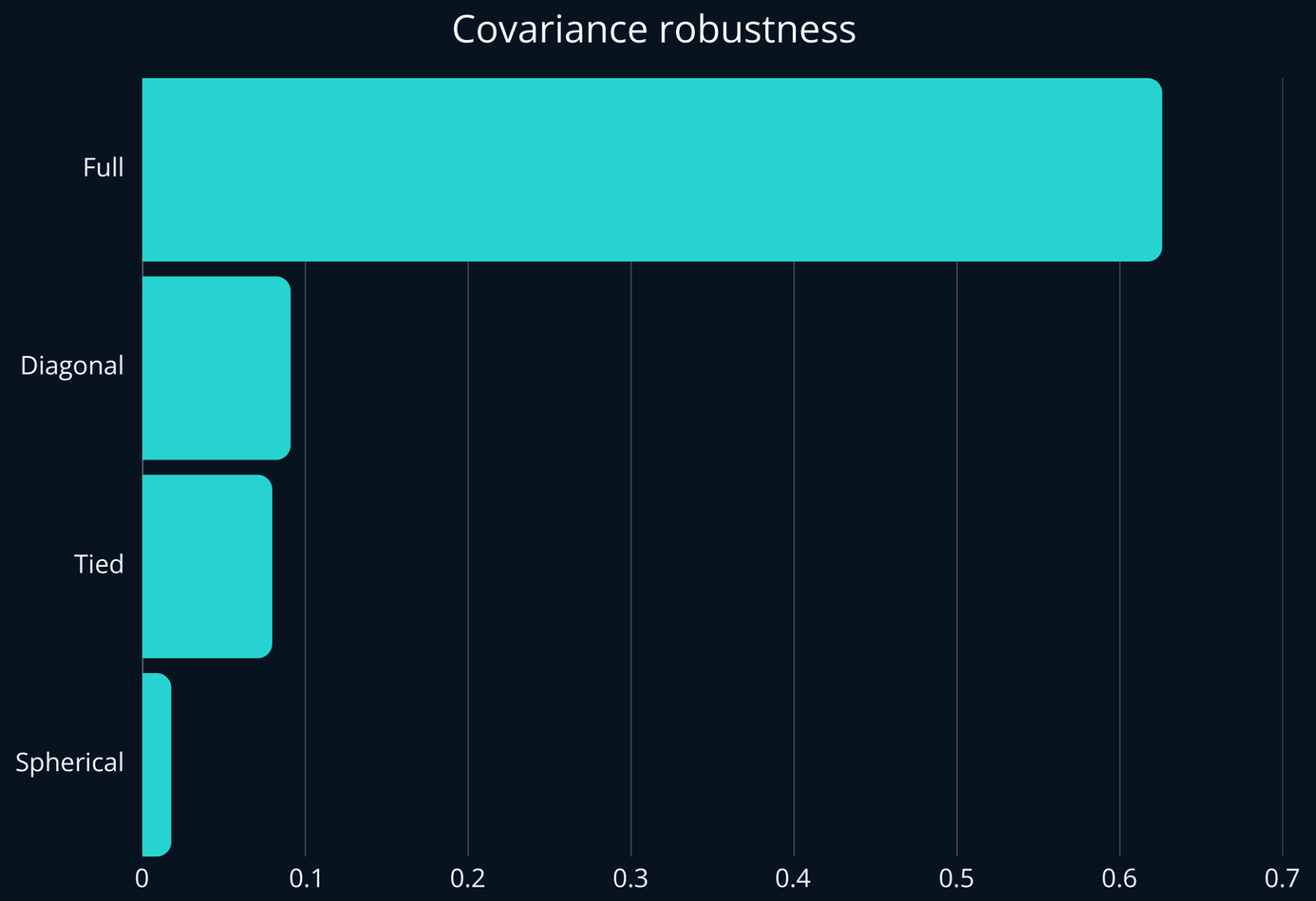
$$s(x) = -\ln p(x|\hat{\theta})$$

Higher score = lower legitimate density

Correlation structure is the decisive choice

BIC selects K = 15 with full covariance

Explored K = 2–15 across full, diagonal, tied and spherical covariance.



Full covariance preserves inter-feature relationships that simpler structures erase.

Held-out ranking is far above random

LOCKED TEST · 42,560 TRANSACTIONS · 52 FRAUD

0.6432

PR-AUC / Average Precision

random baseline 0.0012

0.9512

ROC-AUC

random baseline 0.5000

PR-AUC is primary: it rewards fraud capture while exposing false-alarm cost under extreme imbalance.

Every threshold encodes a workload trade-off

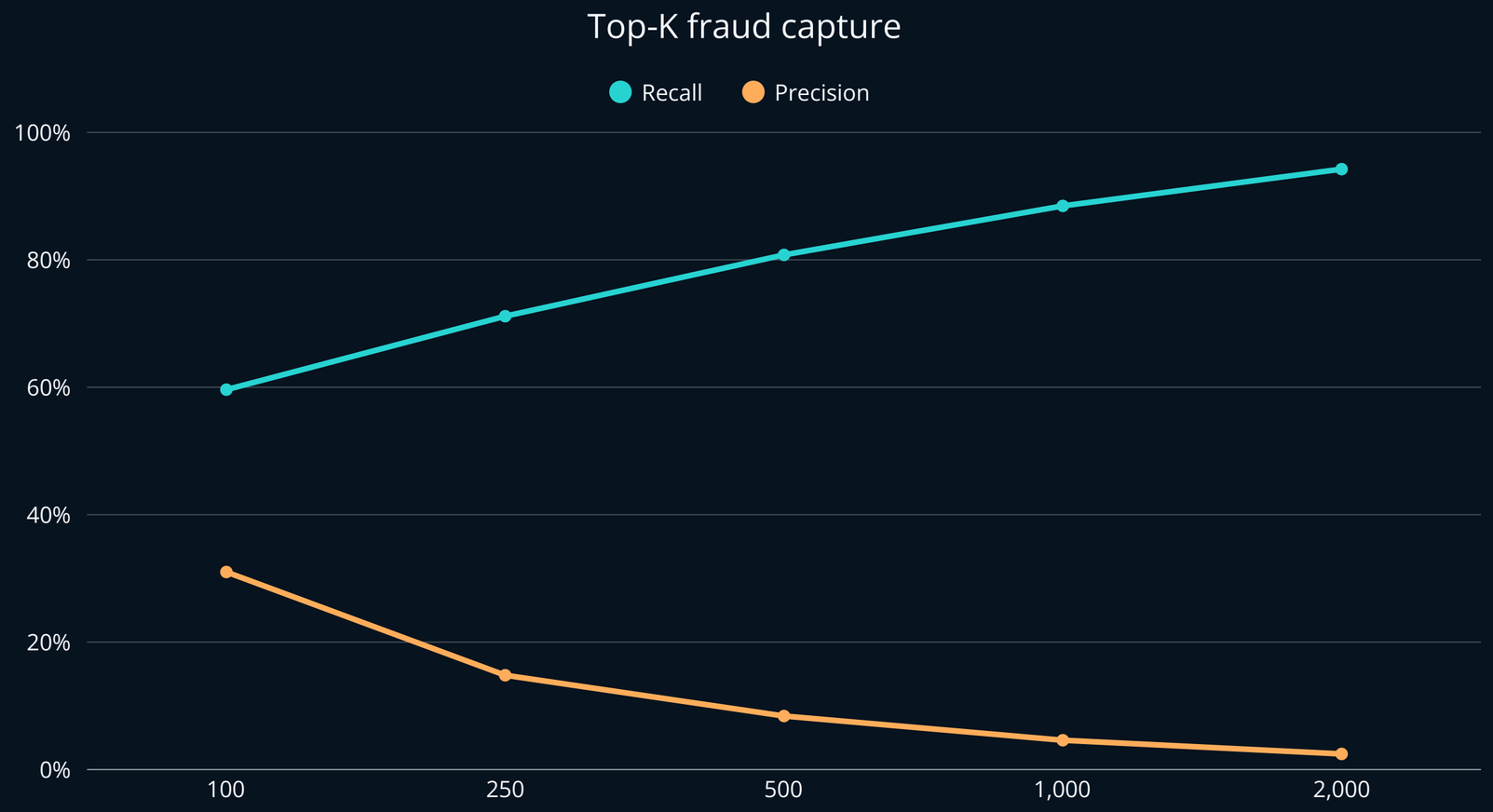
Validation-calibrated thresholds · frozen before held-out test

FPR POLICY	THRESHOLD	RECALL	PRECISION	FLAGGED TX
0.1%	116.88	71.15%	86.05%	43
1.0%	59.50	75.00%	17.33%	225
5.0%	33.37	78.85%	2.87%	1,429

Looser thresholds gain recall slowly—but expand false alarms rapidly. Realized FPR may differ from the target budget.

Review capacity sets the practical operating point

More investigations recover more fraud—at lower precision.



TOP 1,000

46 / 52

fraud detected
88.46% recall

Diminishing returns become visible beyond this queue size.

One score, three operational routes

Offline benchmark policy · not a production guarantee

AUTO-APPROVE

`score < 33.37`

41,131 tx

96.64% · 11 fraud

ANALYST REVIEW

`33.37 ≤ score < 116.88`

1,386 tx

3.26% · 4 fraud

AUTO-BLOCK / HOLD

`score ≥ 116.88`

43 tx

0.10% · 37 fraud

Highest-risk 0.10% delivers 86.05% precision

Phase 1 proves the ranking; Phase 2 expands the model

WHAT WORKED

Legitimate-only GMM delivers useful anomaly ranking. Validation-calibrated thresholds turn scores into capacity-aware decisions.

LIMITATIONS

Gaussian assumptions · high-dimensional density estimation · concept drift · one scalar score may miss fraud subtypes

PHASE 2

Deep autoencoder
→ reconstruction error

Hybrid GMM–autoencoder
→ same leakage-free test

Thank you. Questions?

GitHub github.com/WizardCarp85/Card-Fraud-Triage-ML-DL
Colab colab.research.google.com/drive/1N0ptPhxLE-5Aspw6fHoRRU2ardSHNXj